

MATH 411: Honors Complex Variables

Midterm Exam 1 – Solutions

Fall 2026 | Friday, October 9, 2026

Instructor: Manish Patnaik

Time: 3:00–3:50 PM (50 minutes) **Total: 25 points**

CAB 369

Student name: _____

Student ID: _____

Directions

1. This is a closed-book exam. No notes, textbooks, calculators, or electronic devices may be used.
2. Answer all four questions. Check that your exam has seven pages: this cover; Question 1 on page 2; Question 2 on pages 3–4; Question 3 on page 5; and Question 4 on pages 6–7.
3. Write clearly in the space provided. Show your work and justify your answers. Partial credit may be awarded for meaningful progress.
4. You may use results proved in class. State clearly any result you use and check its hypotheses.
5. If you continue an answer elsewhere in the exam, indicate where the continuation appears and label it with the question and part number.

For instructor use only

Question	Points available	Score
Question 1 (page 2)	7	
Question 2 (pages 3–4)	6	
Question 3 (page 5)	4	
Question 4 (pages 6–7)	8	
Total	25	

1. (7 points) Let $U \subseteq \mathbb{C}$ be an (open) disc and $f = u + iv$ a holomorphic on U . Suppose that

$$u(x, y)^2 + v(x, y)^2 = 2u(x, y) \quad \text{for every } x + iy \in U.$$

Prove that f is constant on U .

Solution

Differentiating with respect to x and y gives

$$2uu_x + 2vv_x = 2u_x, \quad 2uu_y + 2vv_y = 2u_y.$$

Dividing by two, and using the Cauchy–Riemann equations $u_x = v_y$ and $u_y = -v_x$, we obtain the system

$$\begin{pmatrix} u-1 & v \\ v & -(u-1) \end{pmatrix} \begin{pmatrix} u_x \\ v_x \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

The determinant of this system in u_x, v_x is

$$-(u-1)^2 - v^2 = -(u^2 - 2u + 1) - v^2 = -1$$

, so this forces $u_x = v_x = 0$. Thus $f' = u_x + iv_x = 0$ on U . Since U is connected, f is constant.

2. (6 points)

(a) **(2 points)** Determine the radius of convergence of

$$\sum_{n=1}^{\infty} n^2 z^{n^2}.$$

Justify your answer.

Solution

Write the series as $\sum_{m \geq 0} c_m z^m$, where $c_{n^2} = n^2$ and $c_m = 0$ at nonsquare indices. By the Cauchy–Hadamard formula,

$$\frac{1}{R} = \limsup_{m \rightarrow \infty} |c_m|^{1/m} = \lim_{n \rightarrow \infty} (n^2)^{1/n^2} = 1.$$

Hence the radius of convergence is $R = 1$.

- (b) **(4 points)** Let $f(T) = \sum_{n \geq 0} a_n T^n$ and $h(T) = \sum_{n \geq 1} b_n T^n$ be convergent power series, with $h(0) = 0$. Assume that f has radius of convergence $r > 0$ and that $s > 0$ satisfies $\sum_{n \geq 1} |b_n| s^n \leq r$. Let $g = f \circ h$ be the composite power series, i.e.,

$$g(T) = \sum_{n \geq 0} a_n \left(\sum_{k=1}^{\infty} b_k T^k \right)^n.$$

- i. Show that g is a well-defined power series.

Solution

Since $h(0) = 0$, the series $h(T)^n$ has no terms of degree less than n . Thus the coefficient of T^m in $g(T)$ receives contributions only from $n \leq m$. Each such contribution is a finite sum, so every coefficient of g is well-defined.

- ii. Show that g has radius of convergence at least s .

Solution

If $h = 0$, then $g = a_0$ and the claim is immediate. Otherwise, for $0 < t < s$ put $B(t) = \sum_{k \geq 1} |b_k| t^k$. Because h has a nonzero coefficient,

$$B(t) < B(s) \leq r.$$

Let c_m be the coefficient of T^m in g . The triangle inequality for Cauchy products gives

$$\sum_{m \geq 0} |c_m| t^m \leq \sum_{n \geq 0} |a_n| B(t)^n < \infty,$$

since $B(t) < r$. Hence g converges absolutely for every $|T| < s$, so its radius of convergence is at least s .

3. (4 points) Let $U \subseteq \mathbb{C}$ be an (open) disc and let f and g be analytic on U . Suppose that

$$f(z)g(z) = 0 \quad \text{for every } z \in U.$$

Prove that either $f \equiv 0$ on U or $g \equiv 0$ on U .

Solution

If $f \equiv 0$, there is nothing to prove. Otherwise choose $z_0 \in U$ with $f(z_0) \neq 0$. By continuity, f is nonzero on a disc about z_0 , so g vanishes throughout that disc. If g were not identically zero on the connected set U , its zeros would be isolated. Therefore $g \equiv 0$ on U .

4. (8 points) Justify your answers to each part.

(a) (3 points) Compute $\int_C \bar{z} dz$, where C is the counterclockwise-oriented unit circle.

Solution

Parametrize C by $z(t) = e^{it}$, $0 \leq t \leq 2\pi$. Then $\overline{z(t)} = e^{-it}$ and $z'(t) = ie^{it}$, so

$$\int_C \bar{z} dz = \int_0^{2\pi} e^{-it} i e^{it} dt = 2\pi i.$$

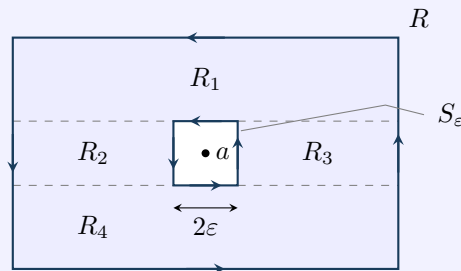
- (b) **(5 points)** Let $R \subseteq \mathbb{C}$ be a closed rectangle, and let a be a point in its interior. Suppose f is continuous on an open neighborhood U of R and holomorphic on $U \setminus \{a\}$. Prove that

$$\int_{\partial R} f(z) dz = 0.$$

Hint: Use Cauchy–Goursat for rectangles and the ML estimate; isolate the point a in a small sub-rectangle.

Solution

Choose a square S_ε centered at a , with side length 2ε and sides parallel to those of R . For sufficiently small ε , it lies in the interior of R . The region between R and S_ε is the union of four closed rectangles that do not contain a .



Dashed edges cancel when the four boundary integrals are added.

The arrows show ∂R and ∂S_ε both counterclockwise.

Cauchy–Goursat applies to each of $R_1, \dots, R_4$. Adding their boundary integrals cancels the shared edges and yields

$$\int_{\partial R} f(z) dz = \int_{\partial S_\varepsilon} f(z) dz,$$

with both displayed boundaries oriented counterclockwise. Since f is continuous on the compact rectangle R , it is bounded there by some M . The ML estimate gives

$$\left| \int_{\partial S_\varepsilon} f(z) dz \right| \leq 8M\varepsilon \longrightarrow 0.$$

The left-hand integral is independent of ε , so it equals zero.